习题10.1:1,2,3(1)(2),4(1)(2),5,6(2)(3).
10.2:1,2,3,4,5,7,9,10
10.3:2(3)(4),3,4,5(1)(5),6,9,11.

10.1

10.1.1

Suppose f is bounded on $[a,b]$. If for any $a' \in (a,b)$, $f \in R[a',b]$, prove that $f \in R[a,b]$ and

$$\lim_{a' \to a^+} \int_{a'}^{b} f \,\mathrm{d}x = \int_{a}^{b} f \,\mathrm{d}x.$$

Proof: Suppose $|f| \leqslant M$ on $[a,b]$. Then

$$-M(a'-a) + \underline{\int_{a'}^{b}} f dx \leqslant \underline{\int_{a}^{b}} f dx \leqslant \overline{\int_{a}^{b}} f dx \leqslant M(a'-a) + \overline{\int_{a'}^{b}} f dx$$

Hence

$$\overline{\int_{a}^{b}} f dx - \underline{\int_{a}^{b}} f dx \leqslant 2M(a'-a) \to 0,$$

so $f \in R[a,b]$, and $\lim_{a' \to a} \int_{a'}^{b} f \,\mathrm{d}x = \int_{a}^{b} f \,\mathrm{d}x$.

10.1.2

Prove that

$$\frac{\pi}{2\sqrt{2}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^4}} < \frac{\pi}{2}.$$

Proof: Note that

$$\frac{\pi}{2\sqrt{2}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{2} \cdot \sqrt{1-x^2}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^4}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^2}} = \frac{\pi}{2}.$$

10.1.3

Calculate
(1)

$$\int_2^{\infty} \frac{\mathrm{d}x}{x(\log x)^2} = \int_{\log 2}^{\infty} \frac{\mathrm{d}y}{y^2} = \frac{1}{\log 2}.$$

(2)

$$\int_1^\infty \frac{\log x}{x^2}\,\mathrm{d}x = \int_0^\infty ye^{-y}\,\mathrm{d}y = \int_0^\infty e^{-y}\,\mathrm{d}y = 1.$$

10.1.4

Calculate
(1)

$$\int_0^1 \frac{x^3}{\sqrt{1-x^2}}\,\mathrm{d}x = \int_0^{\pi/2} \sin^3\theta\,\mathrm{d}\theta = -\int_0^{\pi/2} (1-\cos^2\theta)\,\mathrm{d}\cos\theta = \int_0^1 1-t^2\,\mathrm{d}t = \frac{2}{3}.$$

(2)

$$\int_{-1}^1 \frac{x}{\sqrt{1-x^2}}\,\mathrm{d}x = 0.$$

10.1.5

Determine convergence:
(1)

$$\int_0^\infty |\cos x^2|\,\mathrm{d}x = \int_0^\infty \frac{|\cos y|}{2\sqrt{y}}\,\mathrm{d}y = \infty.$$

(2)

$$\int_0^\infty \frac{\mathrm{d}x}{e^x\sqrt{x}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{x}} + \int_1^\infty \frac{\mathrm{d}x}{x^{3/2}} < \infty.$$

10.1.6

Determine convergence:
(2)

$$\int_0^1 \frac{\log x}{1-x}\,\mathrm{d}x = \int_0^1 \frac{\log(1-x)}{x}\,\mathrm{d}x = -\frac{\pi^2}{6}.$$

(3)

$$\int_0^2 \frac{\mathrm{d}x}{(x-1)^2} = 2\int_0^1 \frac{\mathrm{d}x}{x^2} = \infty.$$

10.2

10.2.1

Determine convergence:
(1)

$$\int_1^\infty \frac{\mathrm{d}x}{(1+x^4)^{1/3}} \leqslant 4^{4/15} \int_1^\infty \frac{\mathrm{d}x}{x^{16/15}} < \infty.$$

(2)

$$\int_0^\infty \frac{\mathrm{d}x}{1+\sqrt{x}} > \frac{1}{2} \int_1^\infty \frac{\mathrm{d}x}{\sqrt{x}} = \infty.$$

(3)

$$\int_0^\infty \frac{x^3}{\sqrt{1+x^7}}\,\mathrm{d}x > \frac{1}{2} \int_1^\infty x^{-1/2}\,\mathrm{d}x = \infty.$$

10.2.2

Determine convergence:

(1)

$$\int_0^1 \frac{\mathrm{d}x}{(1-x^3)^{1/3}} < \int_0^1 \frac{\mathrm{d}x}{(1-x)^{1/3}} = \frac{3}{2}.$$

(2)

$$-\int_0^1 \frac{\mathrm{d}x}{\sqrt{x}\log x} = \int_0^\infty \frac{e^{-x/2}}{x}\,\mathrm{d}x > \int_0^1 \frac{e^{-1/2}}{x}\,\mathrm{d}x = \infty.$$

(3)

$$\int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^4}} < \int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^2}} = \frac{\pi}{2}.$$

10.2.3

Determine convergence and absolute convergence.

(1)

$$\int_1^\infty \frac{\sin\sqrt{x}}{x}\,\mathrm{d}x = 2\int_1^\infty \frac{\sin y}{y}\,\mathrm{d}x$$

converges but not absolutely.

(2)

$$\int_1^\infty \frac{\cos x}{x}\,\mathrm{d}x = \int_1^\infty \frac{\mathrm{d}\sin x}{x} = \sin 1 + \int_1^\infty \frac{\sin x}{x^2}\,\mathrm{d}x$$

converges, but not absolutely.

(3)

$$\int_e^\infty \frac{\log\log x}{\log x}\sin x\,\mathrm{d}x = \int_1^\infty \frac{\log t}{t}\sin e^t\,\mathrm{d}e^t = \int_1^\infty \frac{\log t}{t}\,\mathrm{d}\cos e^t = \int_1^\infty \cos e^t \frac{\log t - 1/t^2}{t^2}\,\mathrm{d}x < \infty.$$

converges.

10.2.4

Determine convergence:
(1)

$$\int_0^\infty \sin x^p \,\mathrm{d}x = \frac{1}{p}\int_0^\infty \frac{\sin y}{y^{1-1/p}}\,\mathrm{d}y$$

converges when $p > -1$ and $1 - 1/p > 0$, i.e. $p > 1$.
(2)

$$\int_0^\infty \frac{\cos x}{x^p}\,\mathrm{d}x$$

converges when $0 < p < 1$.
(3)

$$\int_0^\infty \frac{1-\cos x}{x} e^{-x}\,\mathrm{d}x < \int_0^1 x e^{-x}\,\mathrm{d}x + \int_1^\infty 2e^{-x}\,\mathrm{d}x.$$

converges.

10.2.5

Suppose $f \in C[1,\infty)$. If $\int_1^\infty f^2\,\mathrm{d}x$ converges, prove that $\int_1^\infty \frac{f(x)}{x}\,\mathrm{d}x$ converges.
Proof:

$$\int_1^\infty |\frac{f(x)}{x}|\,\mathrm{d}x \leqslant \frac{1}{2}\int_1^\infty f^2 + x^{-2}\,\mathrm{d}x < \infty.$$

10.2.6

Suppose $f \in R((0,b))$ for all $b > 0$, and $\lim_{x\to\infty} f(x) = 0$, prove that

$$\lim_{b\to\infty}\frac{1}{b}\int_0^b f\,\mathrm{d}x = 0.$$

Proof: Suppose $\varepsilon(N) = \sup_{x>N}|f(x)|$, then $\lim_{N\to\infty}\varepsilon(N) = 0$. Let $F(b) = \frac{1}{b}\int_0^b f\,\mathrm{d}x$, then $|F(b+T)| \leqslant \frac{b}{T}|F(b)| + \varepsilon(b)$. Hence $\limsup_{x\to\infty}|F(x)| \leqslant \varepsilon(b)$ for any $b > 0$, so $F(x) \to 0$ and $\lim_{b\to\infty}\frac{1}{b}\int_0^b f\,\mathrm{d}x = 0$.

10.2.7

Suppose $f \in C[0,\infty)$. If $f > 0$ and $\int_0^\infty \frac{\mathrm{d}x}{f}$ converges, prove that

$$\lim_{b\to\infty}\frac{1}{b^2}\int_0^b f\,\mathrm{d}x = \infty.$$

Proof: Note that

$$\int_b^{2b} f\,\mathrm{d}x \cdot \int_b^{2b} \frac{1}{f}\,\mathrm{d}x \geqslant b^2,$$

so

$$\frac{1}{b^2}\int_0^b f\,\mathrm{d}x > \frac{1}{b^2}\frac{b^2}{4} \Big/ \int_{\frac{2}{b}}^{\infty} \frac{1}{f}\,\mathrm{d}x \to \infty.$$

10.2.9

Suppose γ is the Euler constant, prove that

$$\int_1^{\infty} \frac{1}{[x]} - \frac{1}{x}\,\mathrm{d}x = \gamma.$$

Proof:

$$\int_1^{\infty} \frac{1}{[x]} - \frac{1}{x}\,\mathrm{d}x = \sum_{n=1}^{\infty}\int_n^{n+1} \frac{1}{n} - \frac{1}{x}\,\mathrm{d}x = \sum_{n=1}^{\infty}\frac{1}{n} - \log\frac{n+1}{n} = \lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{n} - \log N = \gamma.$$

10.2.10

Determine convergence of

$$\int_0^{\infty} \frac{\mathrm{d}x}{1+x^4\sin^2 x}$$

Solution:
Note that

$$\int_0^{\infty} \frac{\mathrm{d}x}{1+x^4\sin^2 x} < C + \sum_{n=1}^{\infty}\int_{n\pi}^{(n+1)\pi} \frac{\mathrm{d}x}{1+(n\pi)^4\sin^2 x} < C + \frac{C'}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2} < \infty.$$

where

$$\int_0^{\pi/2} \frac{\mathrm{d}x}{1+a^2\sin^2 x} < \int_0^{\pi/2} \frac{\mathrm{d}x}{1+(2a/\pi)^2x^2} < \frac{\pi}{2a}\cdot\frac{\pi}{2}.$$

Hence the integral converges.

10.3

10.3.2

(2)

$$I_n = \int_0^\infty \frac{\mathrm{d}x}{(x^2+a^2)^n}.$$

Then $I_1 = \frac{\pi}{2a}$, and

$$I_n = -\int_0^\infty x\,\mathrm{d}\frac{1}{(x^2+a^2)^n} = n\int_0^\infty \frac{2x^2}{(x^2+a^2)^{n+1}}\,\mathrm{d}x = 2nI_n - 2na^2 I_{n+1}.$$

Hence $I_{n+1} = \frac{2n-1}{2na^2} I_n$, so $I_{n+1} = \frac{(2n-1)!!}{(2n)!!} a^{-2n-1}\frac{\pi}{2}$.
(3)

$$I_n = \int_0^\infty e^{-ax}\sin^n x\,\mathrm{d}x.$$

Note that $I_0 = \frac{1}{a}, I_1 = \frac{1}{1+a^2}$, and for $n \geqslant 2$,

$$\begin{aligned} I_n &= \int_0^\infty e^{-ax}\sin^n x\,\mathrm{d}x = -\frac{1}{a}\int_0^\infty \sin^n x\,\mathrm{d}e^{-ax} = \frac{1}{a}\int_0^\infty e^{-ax} n\sin^{n-1}x\cos x\,\mathrm{d}x \\ &= -\frac{n}{a^2}\int_0^\infty \sin^{n-1}x\cos x\,\mathrm{d}e^{-ax} = \frac{n}{a^2}\int_0^\infty e^{-ax}((n-1)\sin^{n-2}x\cos^2 x - \sin^n x)\,\mathrm{d}x \\ &= \frac{n}{a^2}((n-1)I_{n-2} - nI_n). \end{aligned}$$

Hence $I_n = \frac{n(n-1)}{n^2+a^2} I_{n-2}$. $I_{2n} = \frac{(2n)!}{a(a^2+2^2)\cdots(a^2+(2n)^2)}$, and $I_{2n+1} = \frac{(2n+1)!}{(1+a^2)\cdots((2n+1)^2+a^2)}$.

10.3.3

Calculate

$$I = \int_0^\infty \frac{\mathrm{d}x}{1+x^6}.$$

Solution: $f(z) = \frac{1}{1+z^6}$ has simple poles at $z = e^{\pm i\frac{\pi}{6}}, e^{\pm i\frac{\pi}{2}}, e^{\pm i\frac{5\pi}{6}}$, and $\mathrm{Res}(f, e^{i\pi/6}) + \mathrm{Res}(f, e^{i\pi/2}) + \mathrm{Res}(f, e^{i5\pi/6}) = -i/3$. Clearly

$$\lim_{R\to\infty}\int_0^{2\pi} \frac{\mathrm{d}Re^{i\theta}}{1+R^6e^{6i\theta}} = 0.$$

Hence

$$I = \frac{1}{2}\int_{-\infty}^\infty \frac{\mathrm{d}x}{1+x^6} = \frac{1}{2}2\pi i\cdot\frac{-i}{3} = \frac{\pi}{3}.$$

10.3.4

Prove that $\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \frac{\pi}{2}$.
Proof:

$$\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \int_0^\infty \int_0^\infty \sin x e^{-xy}\,\mathrm{d}y\,\mathrm{d}x = \int_0^\infty \frac{\mathrm{d}y}{1+y^2} = \frac{\pi}{2}.$$

(Apply Fubini's theorem to $\int_0^R \int_0^\infty \sin x e^{-xy}\,\mathrm{d}y\,\mathrm{d}x$ to make the interchanging valid.)

10.3.5

(1)

$$\int_0^\pi x \log \sin x\,\mathrm{d}x = \pi \int_0^{\pi/2} \log \sin x\,\mathrm{d}x = -\frac{\pi^2}{4}\log 2.$$

where

$$\int_0^{\pi/2} \log \sin x\,\mathrm{d}x = \frac{\pi}{2}\log 2 + \int_0^{\pi/2} \log \sin \frac{x}{2} + \log \cos \frac{x}{2}\,\mathrm{d}x = \frac{\pi}{2}\log 2 + 2\int_0^{\pi/2} \log \sin x\,\mathrm{d}x.$$

(5)

$$\int_0^{\pi/2} \frac{x^2}{\sin^2 x}\,\mathrm{d}x = -\int_0^{\pi/2} x^2\,\mathrm{d}\cot x = 2\int_0^{\pi/2} x\cot x\,\mathrm{d}x = -2\int_0^{\pi/2} \log \sin x\,\mathrm{d}x = \pi \log 2.$$

10.3.6

(1)

$$\int_0^\infty \frac{\mathrm{d}x}{(1+x^2)(1+x^\alpha)} = \int_0^\infty \frac{\mathrm{d}t}{t^2}\frac{1}{(1+t^{-2})(1+t^{-\alpha})} = \int_0^\infty \frac{\mathrm{d}t}{1+t^2}\frac{t^\alpha}{1+t^\alpha}.$$

Hence

$$\int_0^\infty \frac{\mathrm{d}x}{(1+x^2)(1+x^\alpha)} = \frac{1}{2}\int_0^\infty \frac{\mathrm{d}x}{1+x^2} = \frac{\pi}{4}.$$

(2)

$$\int_0^\infty \frac{\log x}{1+x^2}\,\mathrm{d}x = \int_{-\infty}^\infty \frac{xe^x}{1+e^{2x}}\,\mathrm{d}x = 0.$$

Since $\frac{xe^x}{1+e^{2x}}$ is an odd function.

10.3.9

Suppose $f \in C(\mathbb{R})$ and $\lim_{x\to-\infty} f(x) = L$, $\lim_{x\to\infty} f(x) = M$. For $\alpha > 0$, prove that the integral

$$\int_{-\infty}^\infty (f(x+\alpha) - f(x))\,\mathrm{d}x$$

converges, and calculate its value.
Proof:

$$\int_0^\infty f(x+\alpha)-f(x)\,\mathrm{d}x = -\int_0^\alpha f\,\mathrm{d}x + \lim_{x\to\infty}\int_x^{x+\alpha} f\,\mathrm{d}x$$
$$\int_{-\infty}^0 f(x+\alpha)-f(x)\,\mathrm{d}x = \int_0^\alpha f\,\mathrm{d}x - \lim_{x\to-\infty}\int_x^{x+\alpha} f\,\mathrm{d}x.$$

Note that

$$\lim_{x\to\infty}\left|\int_x^{x+\alpha} f - M\,\mathrm{d}x\right| \leqslant \alpha \limsup_{x\to\infty}|f(x)-M| = 0.$$

Hence $\lim_{x\to\infty}\int_x^{x+\alpha} f\,\mathrm{d}x = \alpha M$, and likewise we have

$$\int_{-\infty}^\infty f(x+\alpha)-f(x)\,\mathrm{d}x = \alpha(M-L).$$

10.3.11

Suppose $f\in C(0,\infty)$, $M=\lim_{x\to\infty} f(x)$ exists, and the integral $\int_0^1 \frac{f(x)}{x}\,\mathrm{d}x$ converges. For $\alpha,\beta>0$, prove that

$$\int_0^\infty \frac{f(\alpha x)-f(\beta x)}{x}\,\mathrm{d}x = M\log\frac{\alpha}{\beta}.$$

Proof: WLOG assume $\alpha\geqslant\beta$. Convergence of $\int_0^1 \frac{f(x)}{x}\,\mathrm{d}x$ implies the integral is well defined. Note that

$$\int_0^\infty \frac{f(\alpha x)-f(\beta x)}{x}\,\mathrm{d}x = \lim_{N\to\infty}\int_{\beta N}^{\alpha N}\frac{f(x)}{x}\,\mathrm{d}x + \lim_{\varepsilon\to 0}\int_{\beta\varepsilon}^{\alpha\varepsilon}\frac{f(x)}{x}\,\mathrm{d}x$$

and

$$\lim_{N\to\infty}\int_{\beta N}^{\alpha N}\frac{f(x)}{x}\,\mathrm{d}x = M\log\frac{\alpha}{\beta} + \lim_{N\to\infty}\int_{\beta N}^{\alpha N}\frac{f(x)-M}{x}\,\mathrm{d}x.$$

where

$$\left|\int_{\beta N}^{\alpha N}\frac{f(x)-M}{x}\,\mathrm{d}x\right| \leqslant \log\frac{\alpha}{\beta}\cdot\sup_{x>\beta N}|f(x)-M| \to 0.$$

Also,

$$\lim_{\varepsilon\to 0}\int_{\beta\varepsilon}^{\alpha\varepsilon}\frac{f(x)}{x}\,\mathrm{d}x \to 0 \text{ since } \int_0^1\frac{f(x)}{x}\,\mathrm{d}x \text{ converges.}$$

Hence

$$\int_0^\infty \frac{f(\alpha x)-f(\beta x)}{x}\,\mathrm{d}x = M\log\frac{\alpha}{\beta}.$$